

Expt 7 Spectral Estimation

7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB

PROGRAM

```
clc; clear; close all
```

```
% Given specifications
```

```
Rp = 1;    % Passband Ripple (dB)
```

```
Rs = 20;   % Stopband Attenuation (dB)
```

```
Fp = 1000; % Passband Frequency (Hz)
```

```
Fst = 3000; % Stopband Frequency (Hz)
```

```
Fs = 8000; % Sampling Frequency (Hz)
```

```
% Normalized frequencies
```

```
Wp = Fp / (Fs/2);
```

```
Ws = Fst / (Fs/2);
```

```
% Determine minimum filter order and cutoff frequency
```

```
[N, Wn] = buttord(Wp, Ws, Rp, Rs);
```

```
% Design Butterworth Low-Pass Filter
```

```
[b, a] = butter(N, Wn, 'low');
```

```
% Frequency response
figure;
freqz(b, a);
title('Frequency Response of Butterworth LPF');
```

```
% Impulse response
figure;
[h, n] = impz(b, a, 50);
stem(n, h);
title('Impulse Response');
```

```
% Step response
figure;
stepz(b, a);
title('Step Response');
```

```
% Pole-zero plot
figure;
zplane(b, a);
title('Pole-Zero Plot');
```

```
% Display results
fprintf('Minimum Filter Order (N) = %d\n', N);
fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);
```

OUTPUT



